MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, operational amplifiers. Two product assurance classes and a choice of case outlines and lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit
01	Single operational amplifier - internally compensated
02	Dual operational amplifier - internally compensated
03	Single operational amplifier - externally compensated
04	Single operational amplifier - externally compensated
05	Dual operational amplifier - externally compensated 1/
06	Dual operational amplifier - externally compensated 1/
07	Single operational amplifier, high speed
08	Dual operational amplifier - internally compensated

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat pack
B	F-3 (14-lead, 1/4" x 1/8"), flat pack
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line
D	F-2 (14-lead, 1/4" x 3/8"), flat pack
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line
F	F-5 (16-lead, 1/4" x 3/8"), flat pack
G	A-1 (8-lead), can
H	F-4 (10-lead, 1/4" x 1/4"), flat pack
I	A-2 (10-lead), can
P	D-4 (8-lead, 1/4" x 3/8"), dual-in-line
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1/ Device types 05 and 06 may be monolithic, or they may consist of two separate, independent dice.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	±22 V dc 2/
Input voltage range	±20 V dc 3/
Differential input voltage range	±30 V dc 4/
Input current range	-0.1 mA to +10 mA
Storage temperature range	-65°C to +150°C
Output short-circuit duration	Unlimited 5/
Lead temperature (soldering, 60 seconds)	+300°C
Junction temperature (T_J)	+175°C 6/

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	±5 V dc to 20 V dc
Ambient temperature range	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
14-lead FP	A, B, D	350 mW at T_A = +125°C	60°C/W	140°C/W
Dual-in-line	C, E, P	400 mW at T_A = +125°C	35°C/W	120°C/W
8-lead can	G	330 mW at T_A = +125°C	40°C/W	150°C/W
10-lead can	I	350 mW at T_A = +125°C	40°C/W	140°C/W
10-lead FP	H	330 mW at T_A = +125°C	60°C/W	150°C/W
16-lead FP	F	400 mW at T_A = +125°C	35°C/W	120°C/W
20-terminal square chip carrier package	2	7/ at T_A = +125°C	60°C/W	120°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

2/ Voltages in excess of these may be applied for short-term tests if voltage difference does not exceed 44 volts.
3/ For supply voltages less than ±20 V dc, the absolute maximum input voltage is equal to the supply voltage.
4/ For device types 04, 06, and 07 only, this rating is ±1.0 V unless resistances of 2 kΩ or greater are inserted in series with the inputs to limit current in the input shunt diodes to the maximum allowable value.
5/ Short circuit may be to ground or either supply. Rating applies to +125°C case temperature or +75°C ambient temperature.
6/ For short term test (in the specific burn-in and life test configuration when required and up to 168 hours maximum) T_J = +275°C.
7/ P_D = 102 mW for device 01.
P_D = 75 mW for device 03.
P_D = 149 mW for device 04.

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SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and 1.2.3 herein.

3.2.1 Circuit diagram and terminal connections. The circuit diagram and terminal connections shall be as specified on figure 1.

3.2.2 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request. A typical schematic circuit shall be as specified on figure 2.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating ambient temperature range of -55°C to +125°C and for supply voltages ±5 V dc to ±20 V dc, unless otherwise specified (see table I).

3.4.1 Offset null circuits. Each amplifier having nulling inputs (device types 01, 02, 03, 05, and 07) shall be capable of being nulled 1 mV beyond the specified offset voltage limits for $-55°C \leq T_A \leq +125°C$ using the circuits of figure 3.

3.4.2 Frequency compensation. Device types 01, 02, 07, and 08 shall be free of oscillation when operated in a unity gain non-inverting mode with no external compensation and a source resistance of $\leq$ 10 kΩ, and when operated in any test condition specified herein. Device types 03, 04, 05, and 06 shall be free from oscillation when compensated with a 30 pF capacitor for all gain configurations or a 3 pF capacitor when used with a gain of 10.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

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TABLE I. Electrical performance characteristics.

Characteristics	Symbol	Conditions (figure 4, unless otherwise specified)		Limits -01, -02, -06		-03, -05		-04, -08		-07		Units
				Min	Max	Min	Max	Min	Max	Min	Max	
Input offset voltage	V_{IO}	R_S = 50Ω	T_A = +25°C	-3.0	+3.0	-2.0	+2.0	-0.5	+0.5	-4.0	+4.0	mV
		1/	-55°C ≤ T_A = +125°C	-4.0	+4.0	-3.0	+3.0	-1.0	+1.0	-6.0	+6.0	mV
Input offset voltage temperature sensitivity	ΔV_{IO} / ΔT		T_A = -55°C	-15	+15	-18	+18	-5.0	+5.0	-50	+50	µV/°C
			T_A = +125°C	-15	+15	-15	+15	-5.0	+5.0	-50	+50	µV/°C
Input offset current	I_{IO}	1/	T_A = +25°C	-30	+30	-10	+10	-0.2	+0.2	-40	+40	nA
			-55°C ≤ T_A ≤ +125°C	-70	+70	-20	+20	-0.4	+0.4	-80	+80	nA
Input offset current temperature sensitivity	ΔI_{IO} / ΔT		T_A = -55°C	-500	+500	-200	+200	-2.5	+2.5	-1,000	+1,000	pA/°C
			T_A = +125°C	-200	+200	-100	+100	-2.5	+2.5	-1,000	+1,000	pA/°C
Input bias current	$+I_{IB}$	1/	T_A = +25°C, +125°C	+1.0	+110	+1.0	+75			+1.0	+250	nA
			T_A = -55°C	+1.0	+265	+1.0	+100	-0.1	+3.0	+1.0	+400	nA
			T_A = +25°C					-0.1	+2.0			nA
			T_A = +125°C					-1.0	+2.0			nA
	$-I_{IB}$	1/	T_A = +25°C, +125°C	+1.0	+110	+1.0	+75			+1.0	+250	nA
			T_A = -55°C	+1.0	+265	+1.0	+100	-0.1	+3.0	+1.0	+400	nA
			T_A = +25°C					-0.1	+2.0			nA
			T_A = +125°C					-1.0	+2.0			nA
Power supply rejection ratio	+PSRR	$+V_{CC}$ = 10 V, $-V_{CC}$ = -20 V, R_S = 50Ω	T_A = +25°C	-50	+50	-50	+50	-16	+16	-100	+100	µV/V
			-55°C ≤ T_A ≤ +125°C	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
Power supply rejection ratio	-PSSR	$+V_{CC}$ = 20 V, $-V_{CC}$ = -10 V, R_S = 50Ω	T_A = +25°C	-50	+50	-50	+50	-16	+16	-100	+100	µV/V
			-55°C ≤ T_A ≤ +125°C	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
Input voltage common mode rejection	CMR	$+V_{CC}$ = 20 V, V_{IN} = ±15 V	R_S = 50Ω	80		80		96		80		dB

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristics	Symbol	Conditions (Figure 4, unless otherwise specified)		Limits -01, -02, -08 Min	Limits -01, -02, -08 Max	Limits -03, -05 Min	Limits -03, -05 Max	Limits -04, -06 Min	Limits -04, -06 Max	Limits -07 Min	Limits -07 Max	Units
Adjustment for input offset voltage 2/	$V_{IO\ ADJ(+)}$	$+V_{CC}$ = 20 V		+5		+4		No external ADJ		+7		mV
Adjustment for input offset voltage 2/	$V_{IO\ ADJ(-)}$	$+V_{CC}$ = 20 V			-5		-4	No external ADJ			-7	mV
Output short-circuit current (for positive output)	$I_{OS(+)}$	V_{CC} = 15 V t ≤ 25 ms 3/		-60		-60		-15		-65		mA
Output short-circuit current (for negative output)	$I_{OS(-)}$	$+V_{CC}$ = 15 V t ≤ 25 ms 3/	+25°C ≤ T_A ≤ +125°C		+60		+60		+15		+65	mA
			T_A = -55°C								+80	mA
Supply current	I_{CC}	$+V_{CC}$ = +15 V 4/	T_A = -55°C		+4.2		+3.5		+0.8		+9	mA
			T_A = +25°C		+3.8		+3.0		+0.6		+8	mA
			T_A = +125°C		+3.4		+2.5		+0.6		+7	mA
Output voltage swing (maximum)	V_{OP}	$+V_{CC}$ = 20 V, R_L = 10 kΩ		±16		±16		±16		±17		V
		$+V_{CC}$ = 20 V, R_L = 2 kΩ		±15		±15		Not specified		±16		V
Open loop voltage gain (single ended) 5/	$A_{VS(o)}$	$+V_{CC}$ = 20 V R_L = 2 kΩ, 10 kΩ 6/ V_{OUT} = ±15 V	T_A = +25°C	50		50		80		50		V/mV
			-55°C ≤ T_A ≤ +125°C	25		25		40		32		V/mV
Open loop voltage gain (single ended) 5/	A_{VS}	$+V_{CC}$ = 5 V R_L = 2 kΩ, 10 kΩ 6/ V_{OUT} = ±2 V		10		10		20		10		V/mV
Transient response rise time	$TR_{(tr)}$	See figure 8 7/			+800		+800		+1,000		+40	ns
Transient response overshoot	$TR_{(OS)}$	See figure 8 7/			+25		+25		+50		+50	%
Slew rate 8/	$SR(+)$	V_{IN} = ±5 V; A_V = 1 see figure 9	-55°C ≤ T_A ≤ +25°C	+0.3		9/		+0.05		+50		V/μs
			T_A = +125°C	+0.3		+0.3		+0.05		+40		V/μs
Slew rate 8/	$SR(-)$	V_{IN} = ±5 V; A_V = 1 see figure 9	-55°C ≤ T_A ≤ +25°C	+0.3		9/		+0.05		+80		V/μs
			T_A = +125°C	+0.3		+0.3		+0.05		+40		V/μs

See footnotes at end of table.

MIL-M-38510/1016

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TABLE I. Electrical performance characteristics - Continued.

Characteristics	Symbol	Conditions (Figure 6, unless otherwise specified)		Limits								Units
				-01, -02, -08		-03, -05		-04, -06		-07		
				Min	Max	Min	Max	Min	Max	Min	Max	
Settling time 10/	$t_s(+)$	See figure 5	T_A = +25°C							800		ns
			-55°C ≤ T_A ≤ +125°C							1,200		ns
	$t_s(-)$	See figure 5	T_A = +25°C							800		ns
			-55°C ≤ T_A ≤ +125°C							1,200		ns
Channel separation - device types 02, 05, 06, and 08 only	CS	±V_{CC} = ±20 V; See figure 6 T_A = +25°C		80		80		80				dB
Noise (referred to input) broadband	$NI_{(BB)}$	±V_{CC} = 20 V; bandwidth = 5 kHz T_A = +25°C			15		15		15		25	μV rms
Noise (referred to input) popcorn	$NI_{(PC)}$	±V_{CC} = 20 V; bandwidth = 5 kHz T_A = +25°C			40		80		40		80	μV pk

NOTES:

1/ Tests at common mode V_{CM} = 0, V_{CM} = -15 V, and V_{CM} = +15 V.

2/ $V_{IO(ADJ)}$ is not performed on device type 02, case I only, or on device type 08 for either case.

3/ Continuous short circuit limits will be considerably less than the indicated test limits. Continuous I_{OS} at T_A ≤ +75°C will cause T_J to exceed the maximum of +175°C. For dual devices, I_{OS} is measured one channel at a time.

4/ Value shown is for single devices (01, 03, 08) only. For dual devices (02, 05, 06, and 08) this limit is per single device.

5/ Note that gain is not specified at $V_{IO(ADJ)}$ extremes. Some gain reduction is usually seen at $V_{IO(ADJ)}$ extremes. For closed loop applications (closed loop gain less than 1,000), the open loop tests (A_{VS}) prescribed herein should guarantee a positive, reasonably linear, transfer characteristic. They do not, however, guarantee that the open loop gain is linear, or even positive, over the operating range. If either of these requirements exist (positive open loop gain or open loop gain linearity), they should be specified in the individual procurement document as additional requirements.

6/ R_L = 10 kΩ only for device types 04 and 06.

7/ For transient response tests, C_F = 10 pF for device types 01, 02, 03, 04, 05, 06, and 08. Device type 07, C_F = 47 pF. C_F includes the effects of stray capacitance.

8/ Minimum limit for device 08 is 0.4 V/μs at all temperatures.

9/ Minimum limits for device types 03 and 05 are 0.2 V/μs at -55°C and 0.3 V/μs at both +25°C and +125°C.

10/ Settling time is waived for method 5004, MIL-STD-883 except for device type 07.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1, 2, 3, 4	1, 2, 3, 4
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7, 8, 12, 13	1, 2, 3, 4, 5, 6, 7
Group C end point and group B, class S, electrical parameters (method 5005)	1, 2, 3, and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	Not applicable	8, 12, 13
Group D end point electrical parameters (method 5005)	1, 2, 3	1

1/ PDA applies to subgroup 1 (see 4.2d).

3.7 Marking. Marking shall be in accordance with MIL-M-38510.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 49 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

1. For class S devices: Test condition D using the circuit shown on figure 7.

2. For class B devices: Test condition D using the circuit shown on figure 7 or test condition C using the circuit shown on figure 8, or test condition F using the circuit (or equivalent) shown on figure 9.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 8.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

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d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with either method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883, and herein for groups A, B, C, and inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 9, 10, and 11 shall be omitted.

b. Tests shall be as specified in table II herein.

c. Subgroups 12 and 13 (for device type 07 only) shall be added to table I of method 5005 of MIL-STD-883 for class S only. The class S LTPD for subgroup 12 shall be 5 and for subgroup 13 the class S LTPD shall be 7.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 9. If the alternate burn-in conditions are used, the circuit shown on figure 8 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III, method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

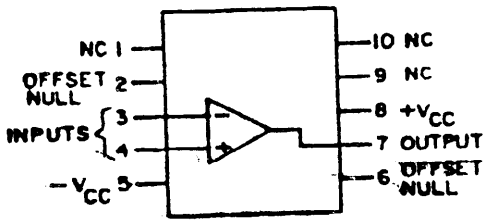
b. Life test for class B (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 7 or test condition C using the circuit shown on figure 8, or test condition F using the circuit shown on figure 9.

c. Subgroups 3, 4, and 5 shall be added to group C inspection and shall consist of subgroups 8, 12, and 13 respectively as specified in table III herein. The LTPD for subgroup 12 shall be 5 for all classes. The LTPD for subgroup 13 shall be 7 for class B.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

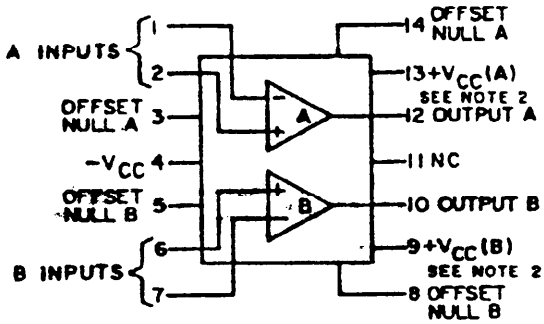
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Device type 01
Case H



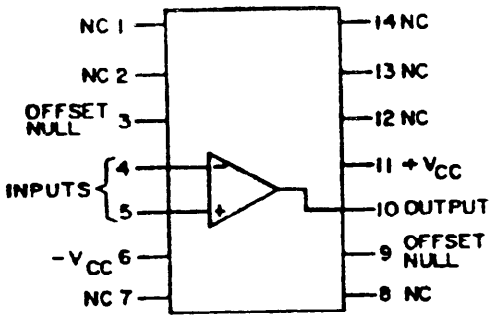
10 Lead Flat Pack (TOP VIEW)

Device type 02
Cases A,B,C and D



14 Lead Flat Pack or Dual-In-Line
(TOP VIEW)

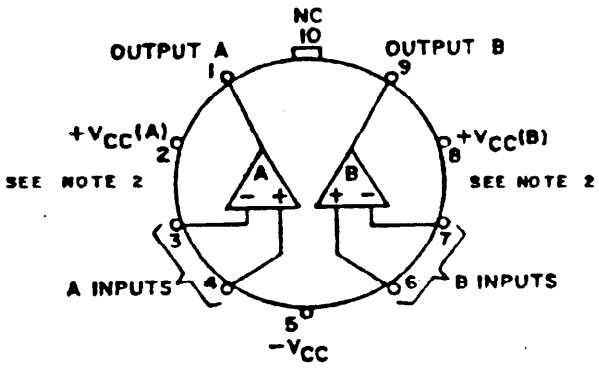
Device type 01
Cases A,B,C and D



14 Lead Flat Pack or Dual-In-Line
(TOP VIEW)

Device type 02

Case 1

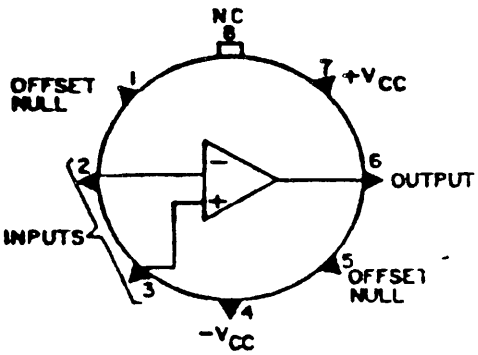


10 Lead Can (TOP VIEW).

FIGURE 1. Terminal connections.

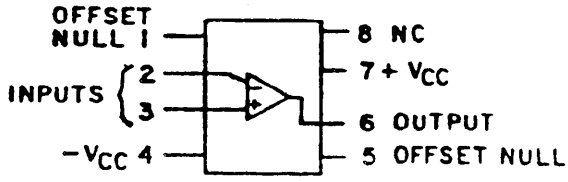
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Device type 01
Case G



8 Lead Can (TOP VIEW)

Device type 01
Case P



8 Lead Dual-In-Line (TOP VIEW)

Device type 01
Case 2

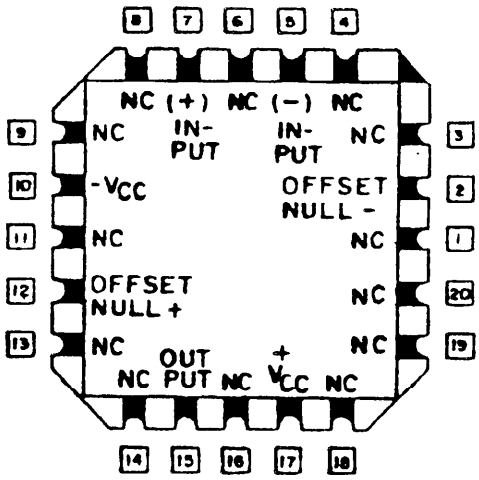
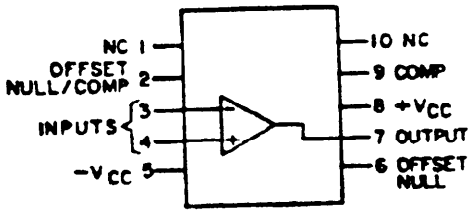


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
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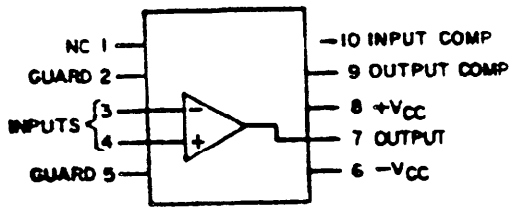
and associated

Device type 03
Case H



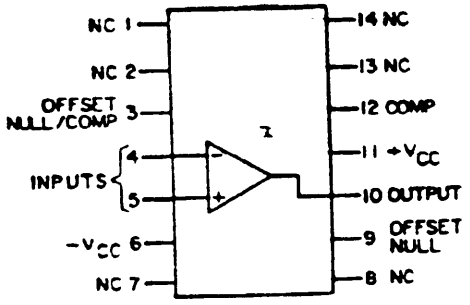
10 Lead Flat Pack (TOP VIEW)

Device type 04
Case H



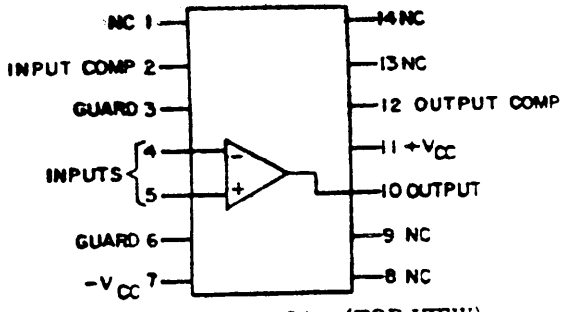
10 Lead Flat Pack (TOP VIEW)

Device type 03
Case C



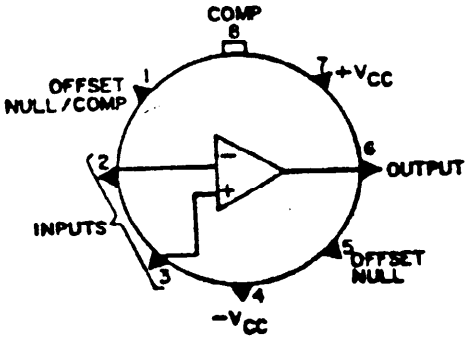
14 Lead Dual-In-Line (TOP VIEW)

Device type 04
Case C



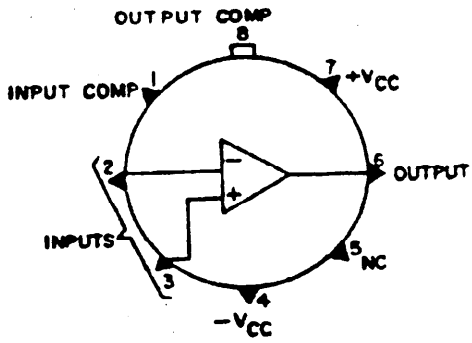
14 Lead Dual-In-Line (TOP VIEW)

Device type 03
Case G



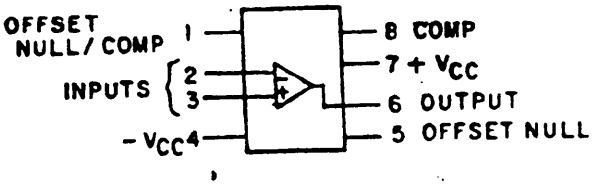
8 Lead Can (TOP VIEW)

Device type 04
Case G



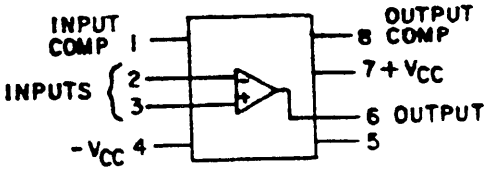
8 Lead Can (TOP VIEW)

Device type 03
Case P



3 Lead Dual-In-Line (TOP VIEW)

Device type 04
Case P

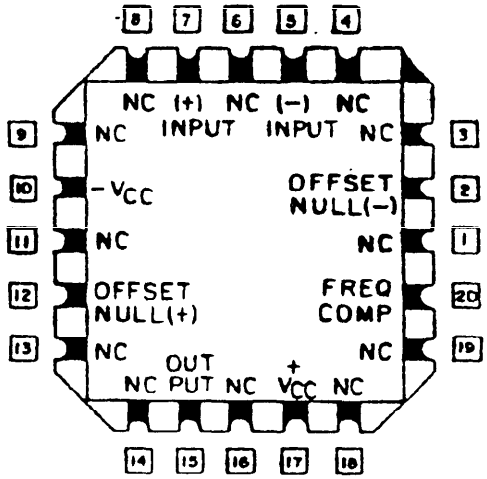


8 Lead Dual-In-Line (TOP VIEW)

FIGURE 1. Terminal connections - Continued.

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Device type 03
Case 2



Device type 04
Case 2

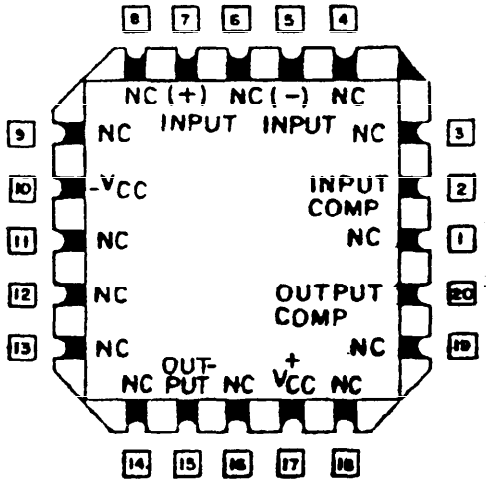
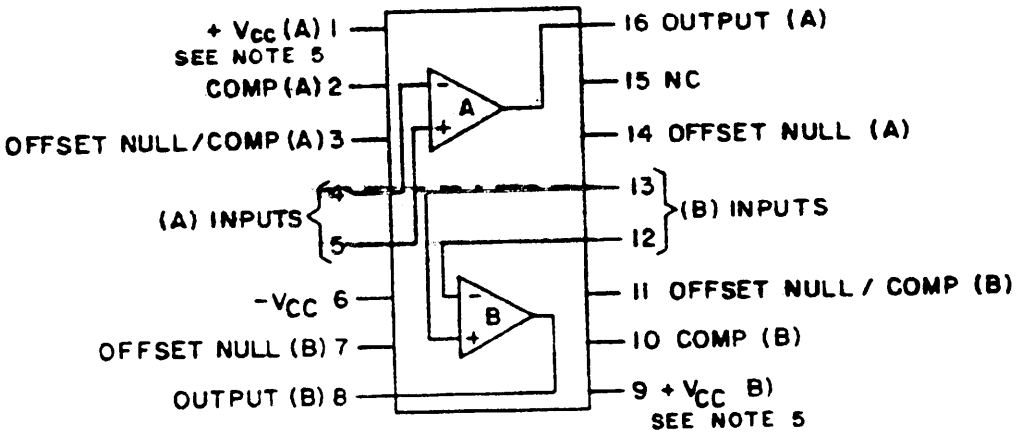


FIGURE 1. Terminal connections - Continued.

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Device type 05

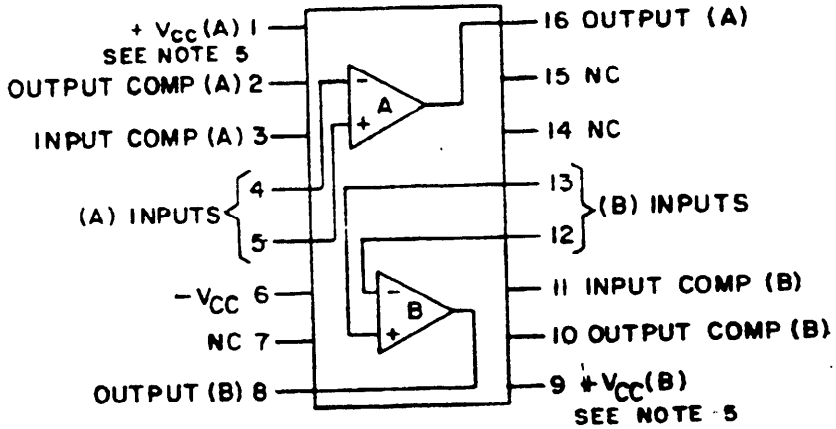
Cases E and F



16 LEAD FLAT-PACK OR DUAL-IN-LINE
(TOP VIEW)

Device type 06

Cases E and F



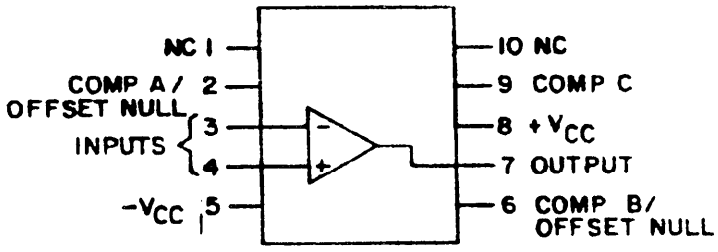
16 LEAD FLAT-PACK AND DUAL-IN-LINE
(TOP VIEW)

FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
Check the source to verify that this is the current version before use.

Device type 07

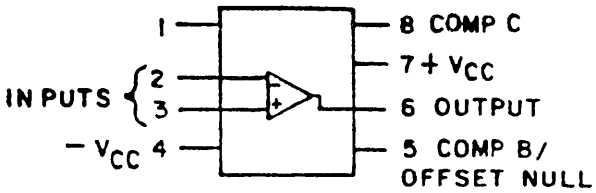
Case H



10 LEAD FLAT PACK (TOP VIEW)

Device type 07

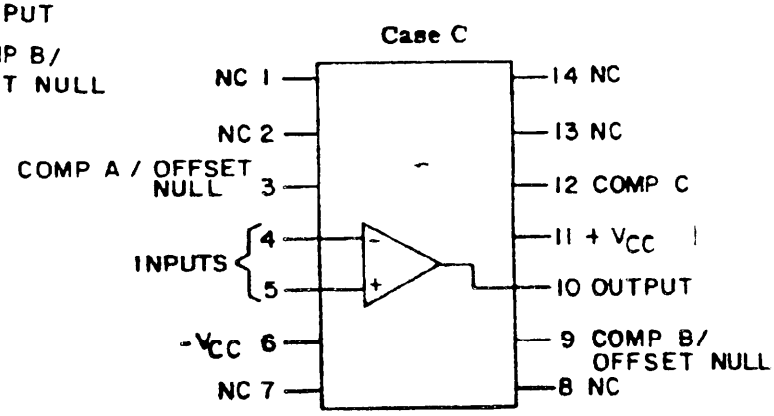
Case P



8 Lead Dual-In-Line (TOP VIEW)

Device type 07

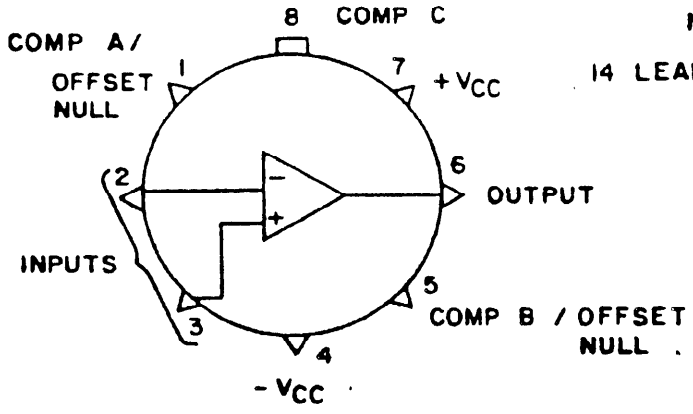
Case C



14 LEAD DUAL-IN-LINE (TOP VIEW)

Device type 07

Case G



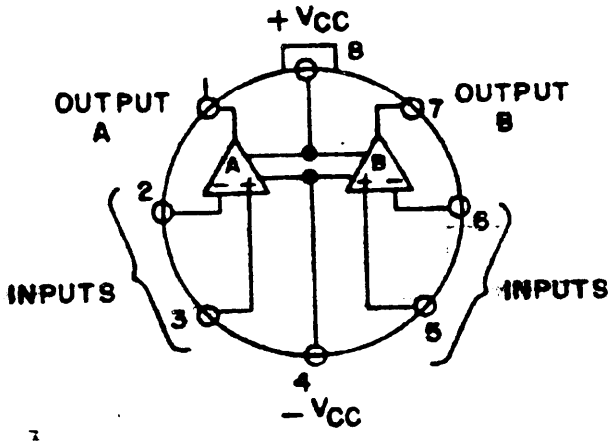
8 LEAD CAN (TOP VIEW)

FIGURE 1. Terminal connections - Continued.

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Device type 08

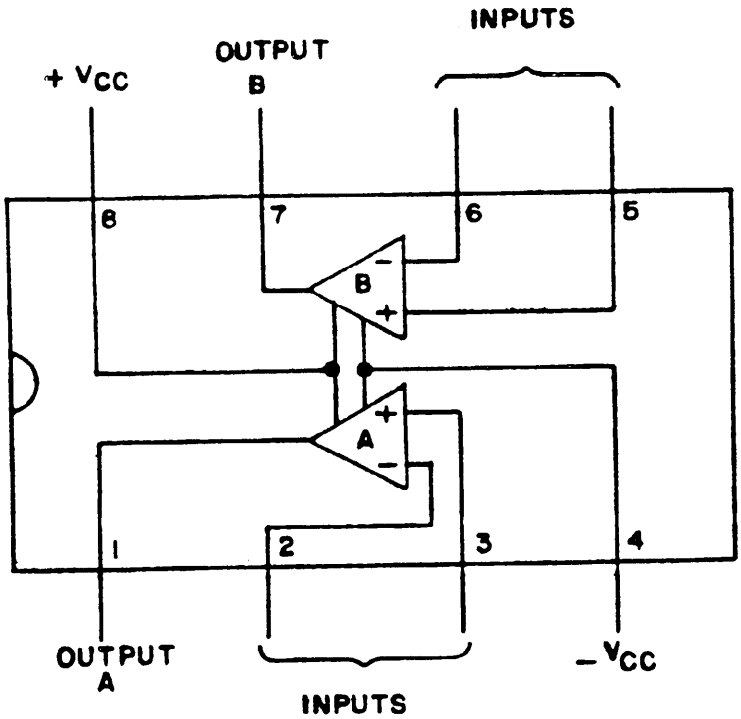
Case G



8-LEAD CAN (TOP VIEW)

Device type 08

Case P

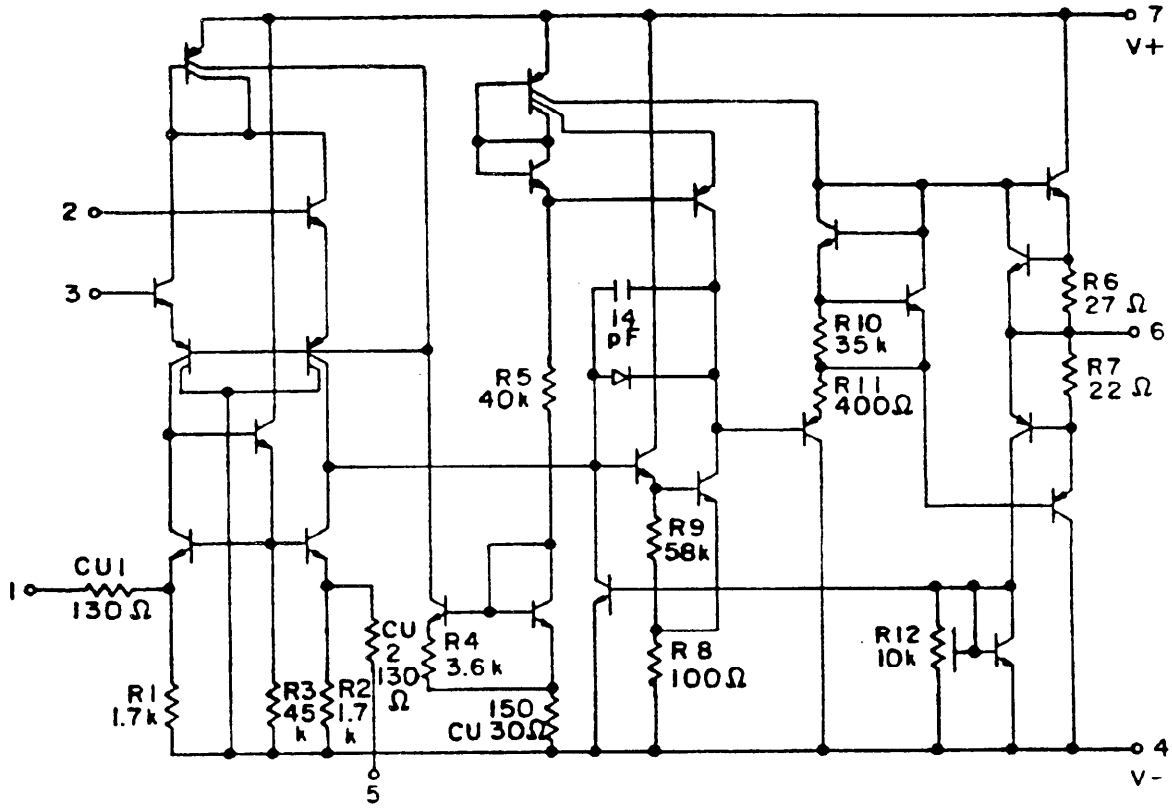


8-LEAD DUAL IN LINE (TOP VIEW)

NOTES:
1. $-V_{CC}$ shall be connected to case of metal packages.
2. Type 02 only, $+V_{CC(A)}$ and $+V_{CC(B)}$ shall be internally connected.
3. +Input is non-inverting input.
4. -Input is inverting input.
5. Types 05 and 06 only, $+V_{CC(A)}$ and $+V_{CC(B)}$ shall not be internally connected. (External connection to the same supply voltage recommended).

FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
Check the source to verify that this is the current version before use.

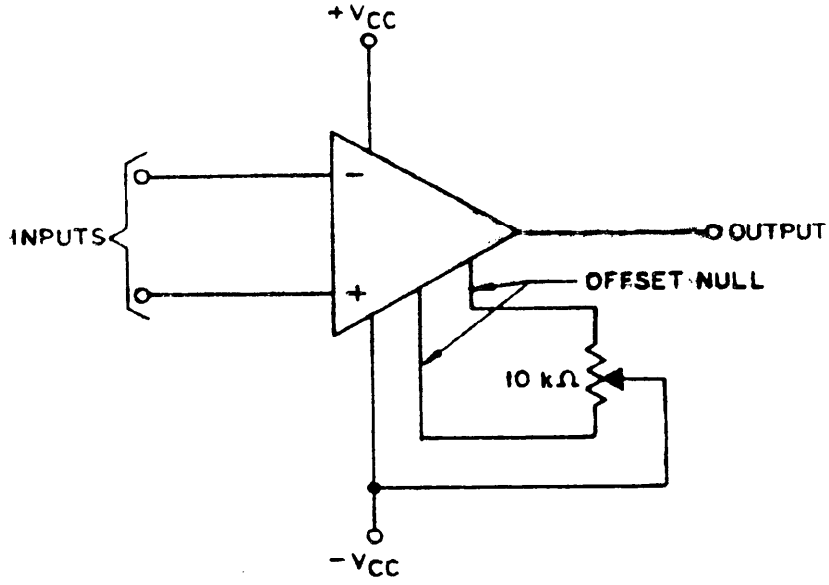


Note: Figure shown represents a single channel.

FIGURE 2. Typical schematic circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
Check the source to verify that this is the current version before use.

Device types 01 and 02



Device types 03 and 05

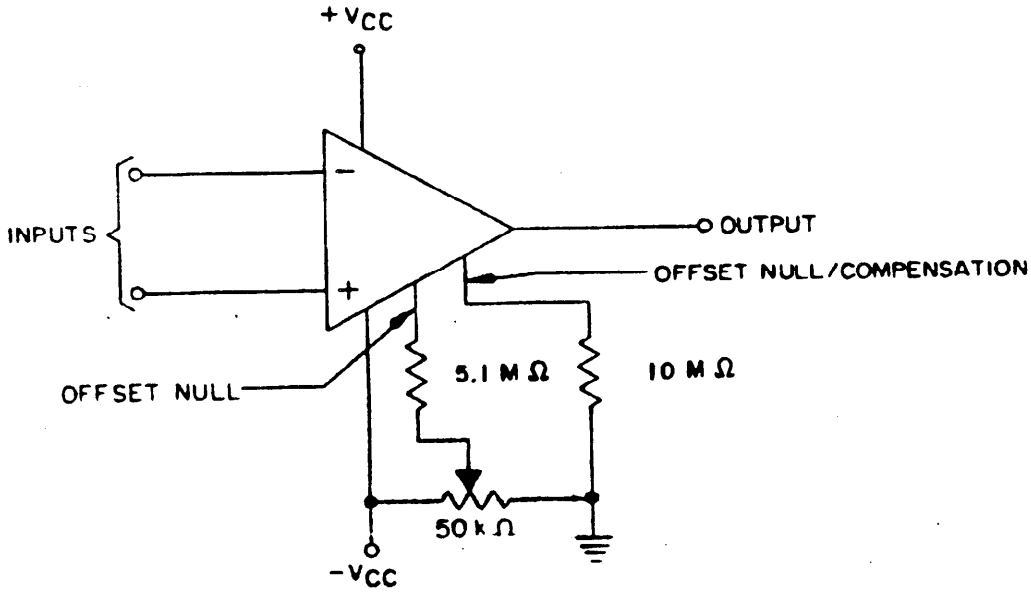


FIGURE 3. Offset null circuits.

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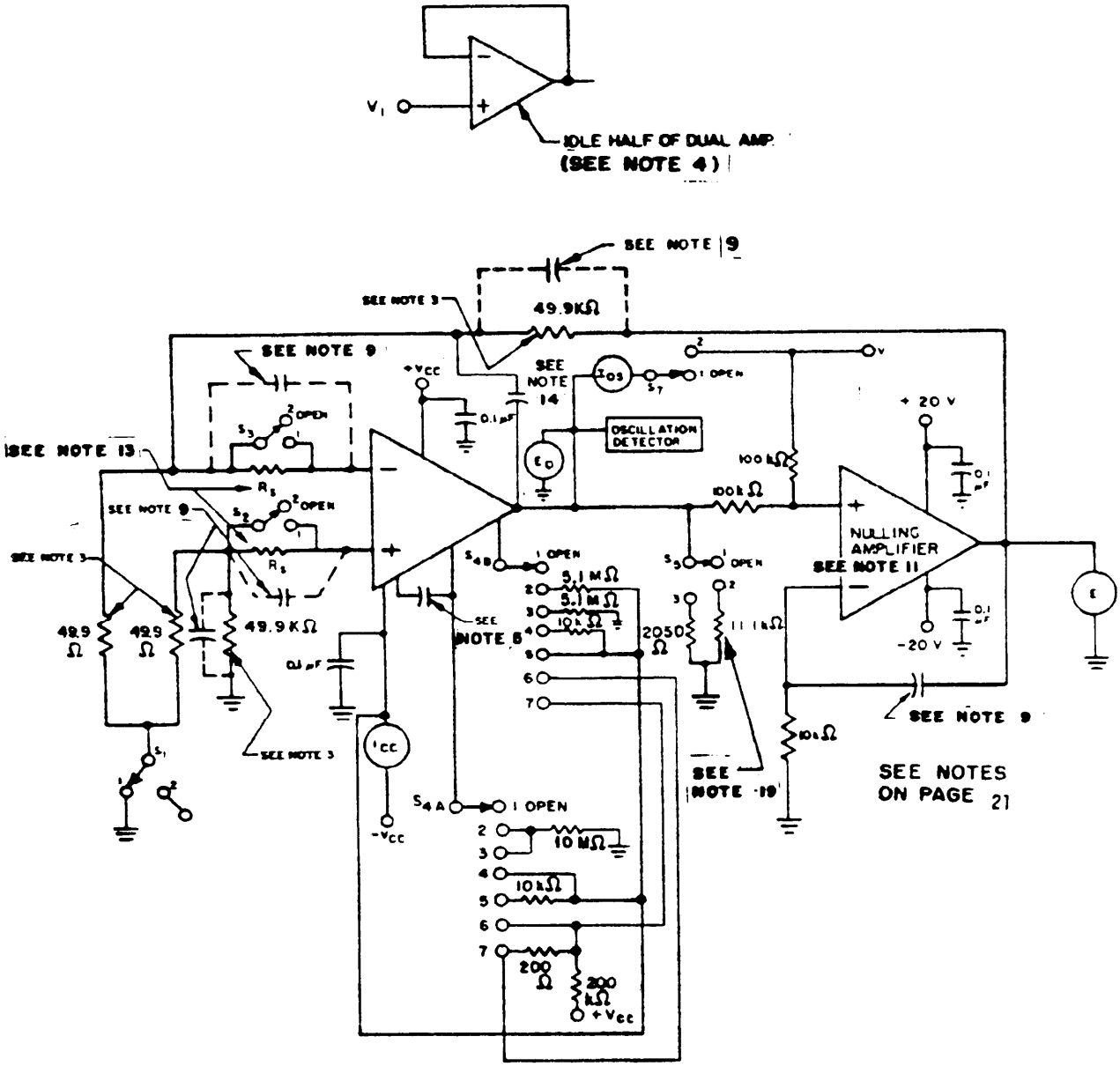


FIGURE 4. Test circuit for static and dynamic tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
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Parameter	Apply (in volts)			Switch positions						Measure		Measured parameter 20/	
	$+V_{CC}$	$-V_{CC}$	V	S_1	S_2	S_3	S_4	S_5	S_6	Value	Units	Equation	Units
V_{IO}	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	E_1 E_2 E_3 E_4	V	$V_{IO} = E_1, E_2, E_3, E_4$	mV
I_{IO}	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	2 2 2 2	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	E_5 E_6 E_7 E_8	V	$I_{IO} = \frac{(E_1-E_5) \times 10^6}{RS}, \frac{(E_2-E_6) \times 10^6}{RS}$ 13/ $\frac{(E_3-E_7) \times 10^6}{RS}, \frac{(E_4-E_8) \times 10^6}{RS}$	nA
$+I_{IB}$	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	E_9 E_{10} E_{11} E_{12}	V	$+I_{IB} = \frac{(E_1-E_9) \times 10^6}{RS}, \frac{(E_2-E_{10}) \times 10^6}{RS},$ 13/ $\frac{(E_3-E_{11}) \times 10^6}{RS}, \frac{(E_4-E_{12}) \times 10^6}{RS}$	nA
$-I_{IB}$	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	1 1 1 1	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	E_{13} E_{14} E_{15} E_{16}	V	$-I_{IB} = \frac{(E_{13}-E_1) \times 10^6}{RS}, \frac{(E_{14}-E_2) \times 10^6}{RS},$ 13/ $\frac{(E_{15}-E_3) \times 10^6}{RS}, \frac{(E_{16}-E_4) \times 10^6}{RS}$	nA
+PSRR	10	-20	5	1	1	1	1	1	1	E_{18} 1/	V	$+PSRR = (E_3-E_{18}) \times 10^2$	μV/V
-PSRR	20	-10	-5	1	1	1	1	1	1	E_{19} 1/	V	$-PSRR = (E_3-E_{19}) \times 10^2$	μV/V
CMR	35 5	-5 -35	-15 +15	1 1	1 1	1 1	1 1	1 1	1 1	E_1 1/ E_2 1/	V V	$CMR = 20 \log \left\lvert \frac{30 \times 10^3}{E_1 - E_2} \right\rvert$	dB
V_{IO} ADJ(+) 7/	20	-20	0	1	1	1	2	1	1	E_{20}	V	V_{IO} ADJ (+) = (E_3-E_{20})	mV
V_{IO} ADJ(-) 7/	20	-20	0	1	1	1	3	1	1	E_{21}	V	V_{IO} ADJ (-) = (E_3-E_{21})	mV
V_{IO} ADJ(+) 6/	20	-20	0	1	1	1	4	1	1	E_{22}	V	V_{IO} ADJ (+) = (E_3-E_{22})	mV
V_{IO} ADJ(-) 6/	20	-20	0	1	1	1	5	1	1	E_{23}	V	V_{IO} ADJ (-) = (E_3-E_{23})	mV
V_{IO} ADJ(+) 15/	20	-20	0	1	1	1	6	1	1	E_{34}	V	V_{IO} ADJ (+) = (E_3-E_{34})	mV
V_{IO} ADJ(-) 15/	20	-20	0	1	1	1	7	1	1	E_{35}	V	V_{IO} ADJ (-) = (E_3-E_{35})	mV

See notes on page 21

FIGURE 4. Test circuit for static and dynamic tests - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T13:16Z
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Parameter	Apply (in volts)			Switch positions						Measure		Measured parameter 20/	
	$+V_{CC}$	$-V_{CC}$	V	S_1	S_2	S_3	S_4	S_5	S_6	Value	Units	Equation	Units
I_{OS}(+output)	15	-15	-15	1	1	1	1	1	2	I_{OS1}	mA	$I_{OS(+)} = I_{OS1}$	mA
I_{OS}(-output)	15	-15	+15	1	1	1	1	1	2	I_{OS2}	mA	$I_{OS(-)} = I_{OS2}$	mA
I_{CC}	15	-15	0	1	1	1	1	1	1	I_{CC}	mA	$I_{CC} = I_{CC}$	mA
$+V_{OP}$ R_L = 10 kΩ	20	-20	-20	1	1	1	1	2	1	$(E_0)_1$	V	$+V_{OP} = (E_0)_1$	V
$-V_{OP}$ R_L = 10 kΩ	20	-20	+20	1	1	1	1	2	1	$(E_0)_2$	V	$-V_{OP} = (E_0)_2$	V
$+V_{OP}$ R_L = 2 kΩ	20	-20	-20	1	1	1	1	3	1	$(E_0)_3$	V	$+V_{OP} = (E_0)_3$	V
$-V_{OP}$ R_L = 2 kΩ	20	-20	+20	1	1	1	1	3	1	$(E_0)_4$	V	$-V_{OP} = (E_0)_4$	V
$+A_{VS}$ 16/ R_L = 2 kΩ	20	-20	-15	1	1	1	1	3	1	E_{24}	V	$+A_{VS} = \frac{15}{E_3 - E_{24}}$	V/mV
$-A_{VS}$ 16/ R_L = 2 kΩ	20	-20	+15	1	1	1	1	3	1	E_{25}	V	$-A_{VS} = \frac{15}{E_{25} - E_3}$	V/mV
A_{VS} 16/ R_L = 2 kΩ	5	-5	-2	1	1	1	1	3	1	E_{26}	V	$A_{VS} = \frac{4}{E_{27} - E_{26}}$	V/mV
	5	-5	+2	1	1	1	1	3	1	E_{27}	V		
$+A_{VS}$ 16/ R_L = 10 kΩ	20	-20	-15	1	1	1	1	2	1	E_{30}	V	$+A_{VS} = \frac{15}{E_3 - E_{30}}$	V/mV
$-A_{VS}$ 16/ R_L = 10 kΩ	20	-20	+15	1	1	1	1	2	1	E_{31}	V	$-A_{VS} = \frac{15}{E_{31} - E_3}$	V/mV
A_{VS} 16/ R_L = 10 kΩ	5	-5	-2	1	1	1	1	2	1	E_{32}	V	$A_{VS} = \frac{4}{E_{33} - E_{32}}$	V/mV
	5	-5	+2	1	1	1	1	2	1	E_{33}	V		
CMR 3/	20	-20	+15	2	1	1	1	1	1	E_{28} 1/	V	$CMR = 20 \log \left\lvert \frac{30 \times 10^3}{E_{28} - E_{29}} \right\rvert$	dB
	20	-20	-15	2	1	1	1	1	1	E_{29} 1/	V		

See notes on page 21

FIGURE 4. Test circuit for static and dynamic tests - Continued.

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unless otherwise specified, the issues of ... in the solicitation.

NOTES:

1/ These voltages in mV shall be measured to four place accuracy to provide required resolution in PSRR and CMR.

2/ Precautions shall be taken to prevent damage to the D.U.T. during insertion into socket and change of switch positions (e.g. disable voltage supplies, current limit $^{+}V_{CC}$, etc.).

3/ If this alternate CMR test is used, these resistors shall be of .01 percent tolerances matched to .001 percent.

4/ Device types 02, 05, and 06 only, test both halves for all tests. The idle half of the dual amplifiers shall be maintained in this configuration where V_I is midway between $^{+}V_{CC}$ and $-V_{CC}$, or the manufacturer has the option to connect the idle half in a V_{IO} configuration such that the inputs are maintained at the same common mode voltage as the D.U.T.

5/ Compensation - device types 03, 04, 05, and 06 only, = 30 pF, device type 07 only = 330 pF (optional).

6/ Device types 01, and 02, cases A, B, C, and D only.

7/ Device types 03 and 05 only.

8/ See figure 10. Noise test circuit.

9/ As required if needed to prevent oscillation. Also, proper wiring procedures shall be followed to prevent oscillation. Loop response and settling time shall be consistent with the test rate such that any value has settled for at least 5 loop time constants before the value is measured.

10/ Adequate settling time shall be allowed such that each parameter has settled to within 5 percent of its final value.

11/ The nulling amplifier is an M38510/10101XXX. Saturation of the nulling amplifier is not allowed on test where E value is measured.

12/ All resistors 0.1 percent tolerance except as noted (note 3).

13/ For device types 01, 02, 07, and 08: R_S = 20 kΩ. For device types 03 and 05: R_S = 100 kΩ. For device types 04 and 06: R_S = 5.0 MΩ.

14/ Device type 07 only, this capacitor = 1,000 pF maximum to prevent oscillations.

15/ Device type 07 only.

16/ To minimize thermal drift, the reference voltages for gain measurements (E3 and E4) shall be taken immediately prior to or after the reading corresponding to device gain (E24, E25, E26, E27, E30, E31, E32 and E33). The gain at R_L = 10 kΩ is essentially the gain at R_L = 2 kΩ is influenced by thermal gradients on the die resulting from power dissipation in the output stage. Hence, it is not linear and may not even be a true approximation of the gain between other than the specified operation points.

17/ Any oscillation greater than 300 mV in amplitude (pk-pk) shall be cause for device failure.

18/ Although switches are depicted as toggle switches, any switching mechanism may be used provided the switching action is achieved without adversely affecting the measurement.

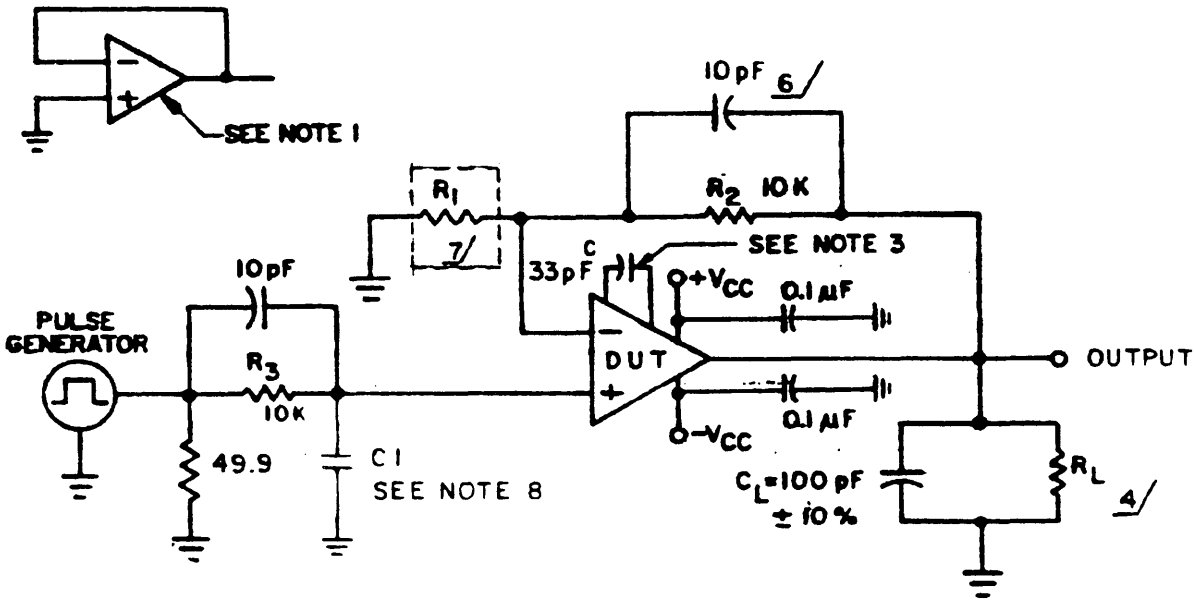
19/ The load resistors (2,050Ω and 11.1 kΩ) yield effective load resistances of 2 kΩ and 10 kΩ, respectively.

20/ The equations take into account both the loop gain of 1,000 and the scale factor multiplier, so that the calculated value is in table III units. Therefore, use measured value/units in the equations, i.e. E1 (volts).

FIGURE 4. Test circuit for static and dynamic tests - Continued.

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Device types 01, 02, 03, 04, 05, 06, and 08



Pulse generator:
1. Rise time $\leq$ 100 ns
2. Repetition rate, 1 kHz (max)

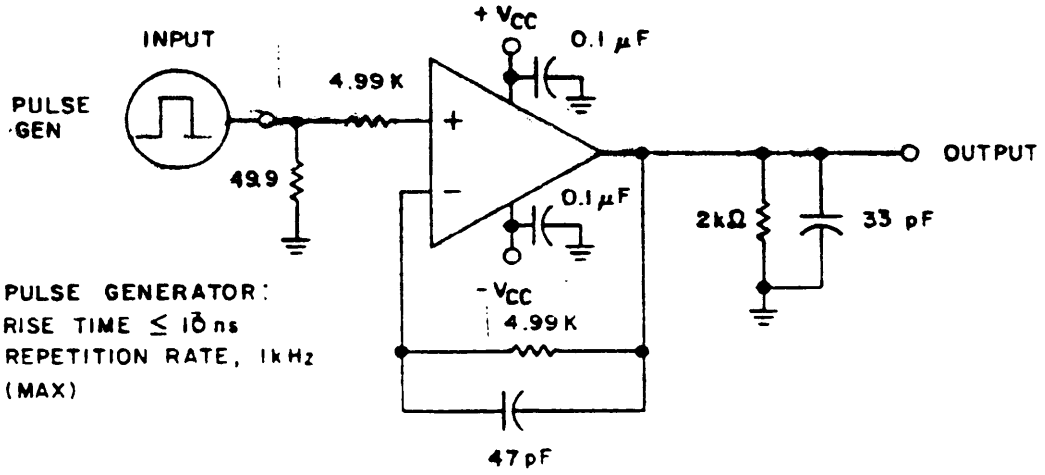
Parameter	Pulse generator	Measure	Equation	Units
t_r Rise time $A_V = 1$	+50 mV amplitude	t (µs) see waveform 1	$t_r = t$	µs
OS Overshoot $A_V = 1$	+50 mV amplitude	ΔV (mV) see waveform 1	$OS = \frac{\Delta V \times 100}{50}$	%
BW Bandwidth $A_V = 1$	+50 mV amplitude	Calculate	$BW = \frac{.35}{t_r\ (\mu s)}$	MHz
SR(+) Slew rate $A_V = 1$	-5 V to +5 V step	ΔV_O (volts), Δt (µs) see waveform 2	$SR(+) = \left\lvert \frac{\Delta V_O(+)}{\Delta t(+)} \right\rvert$	V/µs
SR(-) Slew rate $A_V = 1$	+5 V to -5 V step	ΔV_O (volts), Δt (µs) see waveform 3	$SR(-) = \left\lvert \frac{\Delta V_O(-)}{\Delta t(-)} \right\rvert$	V/µs

See notes on page 24

FIGURE 5. Transient response test circuit.

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Device type 07

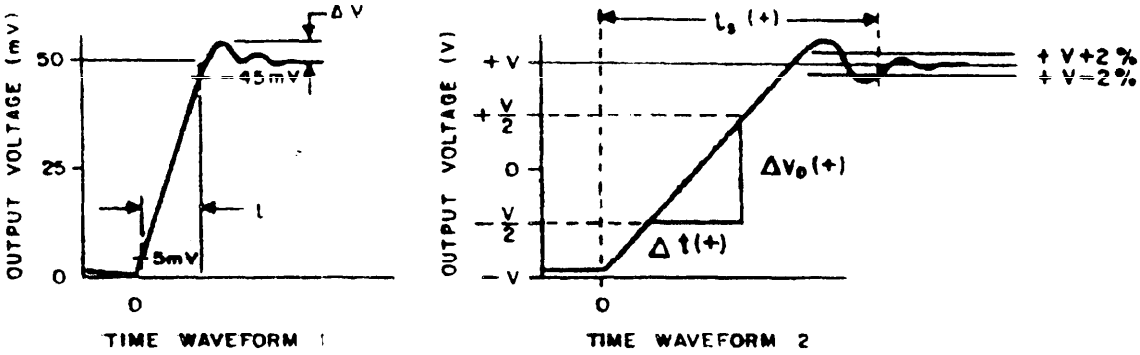


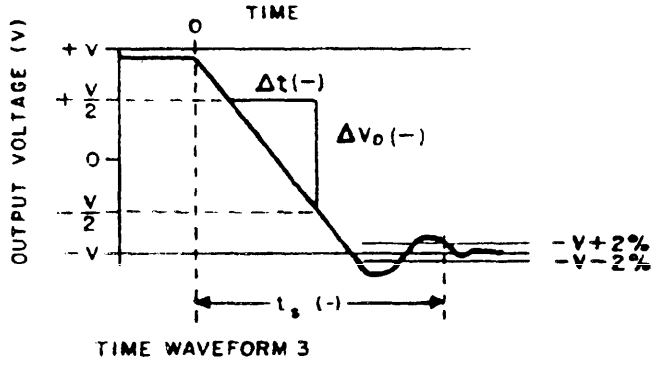
Parameter		Pulse generator	Measure	Equation	Unit
t_r	Rise time	+50 mV amplitude	t (ns) see waveform 1	$t_r = t$	ns
OS	Overshoot	+50 mV amplitude	ΔV (mV) see waveform 1	$OS = \frac{\Delta V \times 100}{50}$	%
BW	Bandwidth	+50 mV amplitude	Calculate	$BW = \frac{.35 \times 10^3}{t_r\ (ns)}$	MHz
SR(+)	Slew rate	-5 V to +5 V step	$\Delta V_O(+)$ volts, Δt(+) ns see waveform 2	$SR(+) = \left\lvert \frac{\Delta V_O(+)}{\Delta t(+) \times 10^{-3}} \right\rvert$	V/μs
SR(-)	Slew rate	+5 V to -5 V step	$\Delta V_O(-)$ volts, Δt(-) ns see waveform 3	$SR(-) = \left\lvert \frac{\Delta V_O(-)}{\Delta t(-) \times 10^{-3}} \right\rvert$	V/μs
$t_s(+)$ 5/	Settling time	Same as for SR(+)	$t_s(+)$. see waveform 2	$t_s(+) = t_s(+)$	ns
$t_s(-)$ 5/	Settling time	Same as for SR(-)	$t_s(-)$ see waveform 3	$t_s(-) = t_s(-)$	ns

See notes on page 24

FIGURE 5. Transient response test circuit - Continued.

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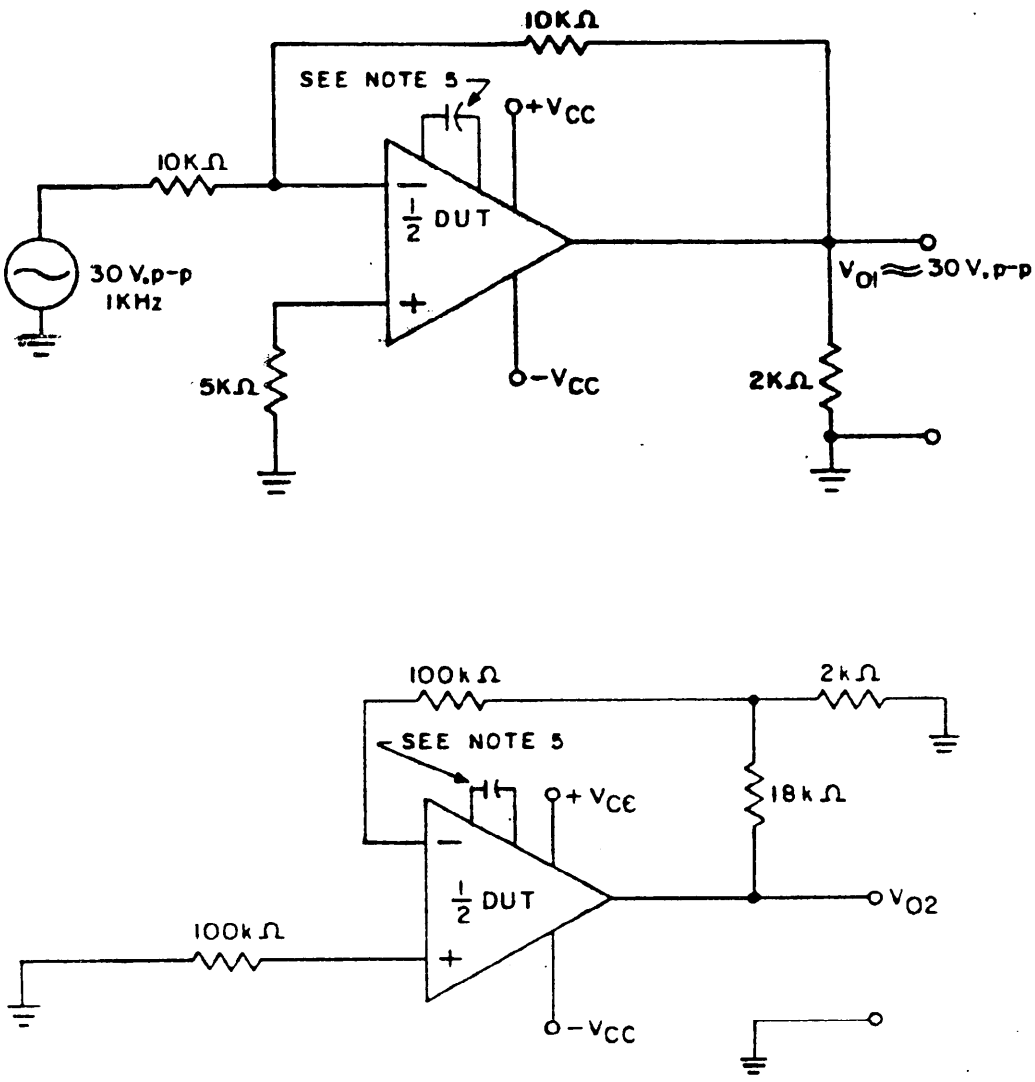




NOTES:
1. Idle half of dual amplifier shall be connected during test of other half.
2. All resistor tolerances are 1 percent, capacitor tolerances are 10 percent and $\pm V_{CC}$ = 20 V.
3. This compensation capacitor is used for device types 03, 04, 05, and 06.
4. For device types 01, 02, 03, 05, and 08, R_L = 2 kΩ; for device types 04 and 06, R_L = 10 kΩ.
5. Settling time is the interval from the beginning of the output response to the point where the output remains within the error band, in this case ±2 percent.
6. CF = 10 pF ±10 percent includes stray capacitance.
7. R1 may be added to the circuit. When R1 is added, its value shall be 10 kΩ. When using R1, the unity gain will increase to 2. To accommodate this change in gain, the pulse generator input voltage shall be halved.
8. C1 may be added to the circuit. When added, it shall be within the range of 0-2 pF.

FIGURE 5. Transient response test circuit - Continued.

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NOTES:
1. *V_{CC} = 20 V.
2. Measure: V_{O2} (volts, p-p) at 1 kHz to accuracy of 0.1 mV or better.
3. Channel separation (dB) referred to input of second channel = $20 \log \left[\frac{V_{O1}}{.1\, V_{O2}} \right]$
4. All resistor tolerances ≤ 1 percent.
5. A 30 pF compensation capacitor is required for device types 05 and 06.

FIGURE 6. Test circuit for channel separation (types 02, 05, 06, and 08 only).

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have passed type C tests does not affect requirements for type B

Device type 07

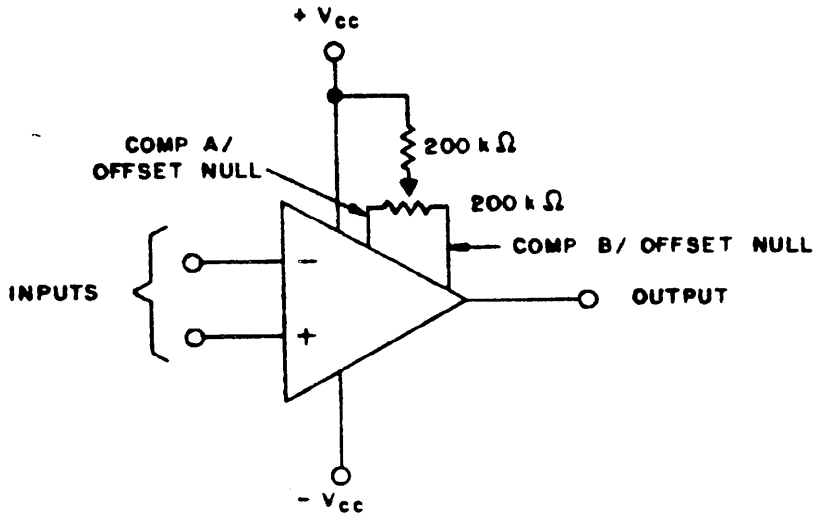
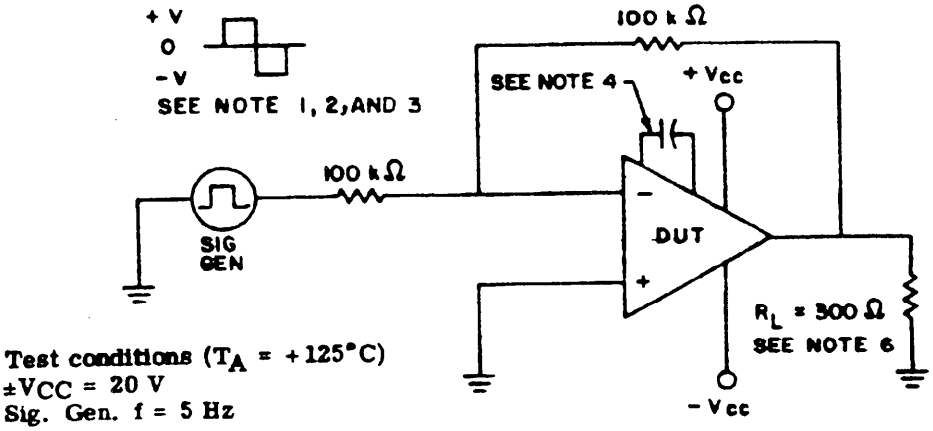


FIGURE 3. Offset null circuits - Continued.

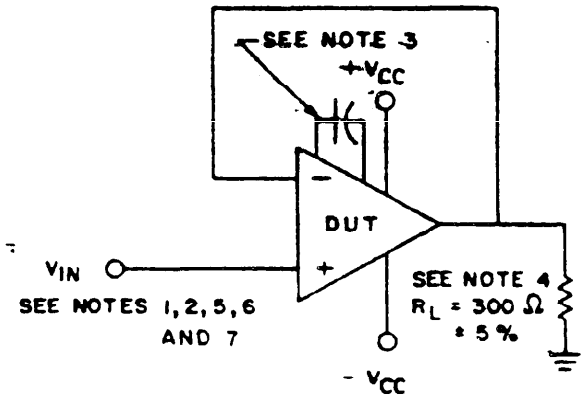


Test conditions (T_A = +125°C)
±VCC = 20 V
Sig. Gen. f = 5 Hz

NOTES:
1. ±V = 3.0 V for device types 01, 03, 04, and 06.
2. ±V = 0.5 V for device types 02 and 08.
3. ±V = 1.0 V for device types 05 and 07.
4. A 30 pF compensation capacitor is required for types 03, 04, 05, and 06.
5. Both halves as above for dual amplifiers, types 02, 05, 06, and 08.
6. For device types 04 and 06, R_L = 1500Ω.
7. All resistor tolerances are ±5 percent and shall not exceed its branded value due to use, heat or age.

FIGURE 7. Test circuit, burn-in and operating life test.

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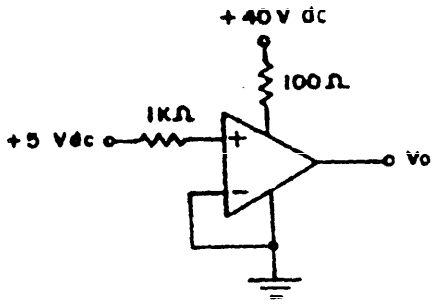
Test conditions (T_A = +125°C)
$\pm V_{CC}$ = 20 V

NOTES:
1. For device types 02 and 08, V_{IN} = 0.5 V dc, and shall be positive into one amplifier and negative into the other amplifier.
2. For device types 01, 03, and 04, V_{IN} = 3.0 V dc.
3. A 30 pF compensation capacitor is required for types 03, 04, 05, and 06.
4. For device types 04 and 06, R_L = 1500Ω. The actual measured value of the resistor selected shall not exceed ± 5 percent of its branded value due to use, heat, or age.
5. For device type 05, V_{IN} = 1.0 V dc, and shall be positive into one amplifier and negative into the other amplifier.
6. For device type 06, V_{IN} = 3.0 V dc, and shall be positive into one amplifier and negative into the other amplifier.
7. For device type 07, V_{IN} = 1.0 V dc.

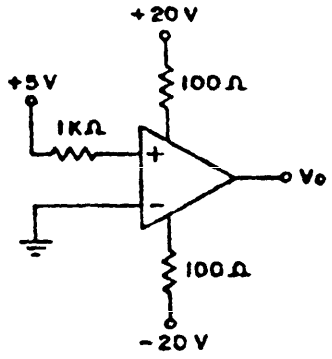
FIGURE 8. Test circuit, burn-in (steady state power and reverse bias) and operating life test.

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Device types 01, 02, 04, 06, 07, and 08



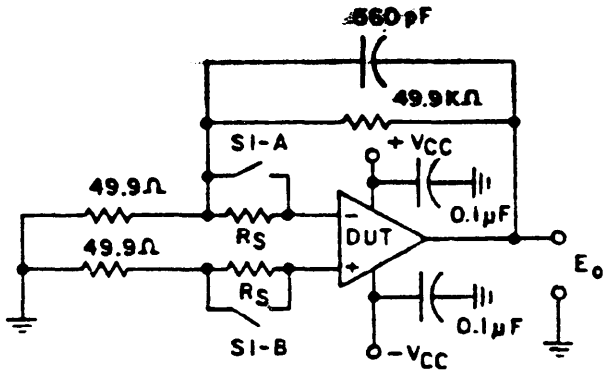
Device types 03 and 05



NOTE:
If accelerated, high temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device during test at temperature. The device is not considered functional under accelerated test conditions. For dual devices both halves are connected as above. The actual measured value of the resistor selected shall not exceed ±5 percent of its branded value due to use, heat, or age.

FIGURE 9. Accelerated burn-in and life test circuit (or equivalent).

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Noise (Referred to Input)	Symbol	S1	Measure VaTue	Measure Units	Measured Equation	Parameter Units
Broadband	$N_{1(BB)}$	Closed	E_o	mV rms	$E_o/1000$	μV rms
Popcorn	$N_{1(PC)}$	Open	E_o	mV pk	$E_o/1000$	μV pk

NOTES:
1. R_S = 20 kΩ for -01, -02, -07, and -08; R_S = 100 kΩ for -03, -04, -05, and -06.
2. E_0 is measured using an RMS voltmeter with a bandwidth of 10 Hz to 5 kHz and a peak detector simultaneously. Monitor the peak test for a minimum of 15 seconds. The loop bandwidth shall be at least 5 kHz.

FIGURE 10. Noise test circuit.

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TABLE III. Group A inspection.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	1/ -01, -02, -08 Limits		1/ -03, -05 Limits		1/ -04, -06 Limits		1/ -07 Limits		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
1 T_A = +25°C	V_{IO}	4001	1	V_{CM} = -15 V 2/	-3.0	+3.0	-2.0	+2.0	-0.5	+0.5	-4.0	+4.0	mV
			2	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			3	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			4	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	I_{IO}		5	V_{CM} = -15 V 2/	-30	+30	-10	+10	-0.2	+0.2	-40	+40	nA
			6	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			7	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			8	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$+I_{IB}$		9	V_{CM} = -15 V 2/	+1.0	110	+1.0	75	-0.1	+2.0	+1.0	250	nA
			10	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			11	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			12	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$-I_{IB}$		13	V_{CM} = -15 V 2/	+1.0	110	+1.0	75	-0.1	+2.0	+1.0	250	nA
			14	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			15	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			16	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$+PSRR$	4003	17	$+V_{CC}$ = 10 V, $-V_{CC}$ = -20 V	-50	+50	-50	+50	-16	+16	-100	+100	μV/V
	$-PSRR$		18	$+V_{CC}$ = 20 V, $-V_{CC}$ = -10 V	-50	+50	-50	+50	-16	+16	-100	+100	μV/V
	CMR		19	V_{CM} = +15 V	80		80		96		80		dB
	V_{IO} ADJ(+)		20	3/	+5		+4				+7		mV
	V_{IO} ADJ(-)		21	3/		-5		-4				-7	mV
	I_{OS}(+) 4/	3011	22	$+V_{CC}$ = +15 V, t ≤ 25 mS	-60		-60		-15		-65		mA
	I_{OS}(-) 4/		23	$+V_{CC}$ = +15 V, t ≤ 25 mS		+60		+60		+15		+65	mA
	I_{CC}	4005	24	$+V_{CC}$ = +15 V 5/		3.8		3.0		0.6		8	mA

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	-01, -02, -08 Limits 1/		-03, -05 Limits 1/		-04, -06 Limits 1/		-07 Limits 1/		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
2 T_A = +125°C	V_{IO}	4001	25	V_{CM} = -15 V 2/	-4.0	+4.0	-3.0	+3.0	-1.0	+1.0	-6.0	+6.0	mV
			26	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			27	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			28	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$\frac{\Delta V_{IO}}{\Delta T}$ 6/		29	$\frac{\Delta V_{IO}}{\Delta T}$ = $\frac{V_{IO}\text{ (test 27)} - V_{IO}\text{ (test 3)}}{}$ x 10	-15	+15	-15	+15	-5.0	+5.0	-50	+50	µV/°C
	I_{IO}		30	V_{CM} = -15 V 2/	-30	+30	-10	+10	-0.4	+0.4	-80	+80	nA
			31	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			32	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			33	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$\frac{\Delta I_{IO}}{\Delta T}$ 6/		34	$\frac{\Delta I_{IO}}{\Delta T}$ = $\frac{I_{IO}\text{ (test 32)} - I_{IO}\text{ (test 7)}}{}$ x 10	-200	+200	-100	+100	-2.5	+2.5	-1,000	+1,000	pA/°C
	$+I_{IB}$		35	V_{CM} = -15 V 2/	+1.0	110	+1.0	75	-1.0	+2.0	+1.0	290	nA
			36	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			37	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			38	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$-I_{IB}$		39	V_{CM} = -15 V 2/	"	"	"	"	"	"	"	"	"
			40	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			41	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			42	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$I_{OS(+)}$ 6/	3011	43	$+V_{CC}$ = +15 V, t ≤ 25 mS	-60		-60		-15		-66		mA
	$I_{OS(-)}$ 6/		44	$+V_{CC}$ = +15 V, t ≤ 25 mS		60		60		15		66	mA
	I_{CC}	4005	45	$+V_{CC}$ = +15 V 5/		3.4		2.6		0.6		7	mA
	+PSRR	4003	46	$+V_{CC}$ = 10 V, $-V_{CC}$ = -20 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
	-PSRR		47	$+V_{CC}$ = 20 V, $-V_{CC}$ = -10 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V

See Footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	-01,-02,-08 Limits 1/ Min	-01,-02,-08 Limits 1/ Max	-03, -05 Limits 1/ Min	-03, -05 Limits 1/ Max	-04, -06 Limits 1/ Min	-04, -06 Limits 1/ Max	-07 Limits 1/ Min	-07 Limits 1/ Max	Unit
2 T_A = +125°C	CMR	4003	48	V_{CM} = +15 V	80		80		96		80		dB
	V_{IO} ADJ(+)		49	3/	+5		+4				+7		mV
	V_{IO} ADJ(-)		50	3/		-5		-4				-7	mV
3 T_A = -55°C	V_{IO}	4001	51	V_{CM} = -15 V 2/	-4.0	+4.0	-3.0	+3.0	-1.0	+1.0	-6.0	+6.0	mV
			52	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			53	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			54	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$\frac{\Delta V_{IO}}{\Delta T}$ 6/		55	$\Delta V_{IO}/\Delta T = [V_{IO}$ (test 3) - V_{IO} (test 53)] x 12.5	-15	+15	-10	+10	-5.0	+5.0	-50	+50	µV/°C
	I_{IO}		56	V_{CM} = -15 V 2/	-70	+70	-20	+20	-0.4	+0.4	-80	+80	nA
			57	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			58	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			59	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$\frac{\Delta I_{IO}}{\Delta T}$ 6/		60	$\Delta I_{IO}/\Delta T = [I_{IO}$ (test 7) - I_{IO} (test 58)] x 12.5	-500	+500	-200	+200	-2.5	+2.5	-1,000	+1,000	pA/°C
	$+I_{IB}$		61	V_{CM} = -15 V 2/	+1.0	265	+1.0	100	-0.1	3.0	+1.0	400	nA
			62	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			63	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			64	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
	$-I_{IB}$		65	V_{CM} = -15 V 2/	"	"	"	"	"	"	"	"	"
			66	V_{CM} = +15 V 2/	"	"	"	"	"	"	"	"	"
			67	V_{CM} = 0 V	"	"	"	"	"	"	"	"	"
			68	$+V_{CC}$ = +5 V, V_{CM} = 0 V	"	"	"	"	"	"	"	"	"

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	-01,-02,-08 Limits 1/		-03, -05 Limits 1/		-04, -06 Limits 1/		-07 Limits 1/		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
3 T_A = -55°C	I_{OS}(+) 4/	3011	69	$+V_{CC}$ = +15 V, t ≤ 25 mS	-60		-60		-15		-65		mA
	I_{OS}(-) 4/		70	$+V_{CC}$ = +15 V, t ≤ 25 mS		60		60		15		80	mA
	I_{CC}	4005	71	$+V_{CC}$ = +15 V 5/		4.2		3.5		0.8		9	mA
	+PSRR	4003	72	$+V_{CC}$ = 10 V, $-V_{CC}$ = -20 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
	-PSRR		73	$+V_{CC}$ = 20 V, $-V_{CC}$ = -10 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
	CMR		74	V_{CM} = +15 V	80		80		96		80		dB
	V_{IO} ADJ(+)		75	3/	+5		+4				+7		mV
	V_{IO} ADJ(-)		76	3/		-5		-4				-7	mV
4 T_A = +25°C	$+V_{OP}$	4004	77	R_L = 10 kΩ	+16		+16		+16		+17		V
	$-V_{OP}$		78	R_L = 10 kΩ		-16		-16		-16		-17	
	$+V_{OP}$		79	R_L = 2 kΩ	+15		+15				+16		
	$-V_{OP}$		80	R_L = 2 kΩ		-15		-15				-16	
	$+A_{VS}$		81	R_L = 2 kΩ, V_{OUT} = -15 V	50		50				50		V/mV
	$-A_{VS}$		82	R_L = 2 kΩ, V_{OUT} = +15 V	50		50				50		
	$+A_{VS}$		83	R_L = 10 kΩ, V_{OUT} = -15 V	50		50		80		50		
	$-A_{VS}$		84	R_L = 10 kΩ, V_{OUT} = +15 V	50		50		80		50		
	A_{VS}		85	R_L = 2 kΩ, $\pm V_{CC}$ = ±5 V dc, V_{OUT} = ±2 V	10		10				10		
	A_{VS}		86	R_L = 2 kΩ, $\pm V_{CC}$ = ±5 V dc, V_{OUT} = ±2 V	10		10		20		10		
5 T_A = +125°C	$+V_{OP}$		87	R_L = 10 kΩ	+16		+16		+16		+17		V
	$-V_{OP}$		88	R_L = 10 kΩ		-16		-16		-16		-17	
	$+V_{OP}$		89	R_L = 2 kΩ	+15		+15				+16		
	$-V_{OP}$		90	R_L = 2 kΩ		-15		-15				-16	

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	-01, -02, -08 Limits 1/ Min	-01, -02, -08 Limits 1/ Max	-01, -05 Limits 1/ Min	-01, -05 Limits 1/ Max	-04, -06 Limits 1/ Min	-04, -06 Limits 1/ Max	-07 Limits 1/ Min	-07 Limits 1/ Max	Unit
5 T_A = +125°C	$+A_{VS}$		91	R_L = 2 kΩ, V_{OUT} = -15 V	25		25				32		V/mV
	$-A_{VS}$		92	R_L = 2 kΩ, V_{OUT} = +15 V	25		25		40		32		
	$+A_{VS}$		93	R_L = 10 kΩ, V_{OUT} = -15 V	25		25		40		32		
	$-A_{VS}$		94	R_L = 10 kΩ, V_{OUT} = +15 V	25		25				10		
	$+A_{VS}$		95	R_L = 2 kΩ, $+V_{CC}$ = +5 V dc, V_{OUT} = +2 V	10		10		20		10		
	$-A_{VS}$		96	R_L = 2 kΩ, $+V_{CC}$ = +5 V dc, V_{OUT} = +2 V	10		.16		.16		.17		V
6 T_A = -55°C	$+V_{OP}$		97	R_L = 10 kΩ		.16		-.16		-.16		-.17	
	$-V_{OP}$		98	R_L = 10 kΩ		.15		-.15		-.15		.16	
	$+V_{OP}$		99	R_L = 2 kΩ			-.15		-.15			-.16	
	$-V_{OP}$		100	R_L = 2 kΩ	25		25				32		V/mV
	$+A_{VS}$		101	R_L = 2 kΩ, V_{OUT} = -15 V	25		25		40		32		
	$-A_{VS}$		102	R_L = 2 kΩ, V_{OUT} = +15 V	25		25		40		32		
	$+A_{VS}$		103	R_L = 10 kΩ, V_{OUT} = -15 V	25		25		40		32		
	$-A_{VS}$		104	R_L = 10 kΩ, V_{OUT} = +15 V	10		10		20		10		
	A_{VS}		105	R_L = 2 kΩ, $+V_{CC}$ = +5 V dc, V_{OUT} = +2 V	10		10		20		10		
	A_{VS}		106	R_L = 2 kΩ, $+V_{CC}$ = +5 V dc, V_{OUT} = +2 V	10				600	11,000			ns
7 T_A = +25°C	$T(t_r)$		107	Figure 5	800		25		25	50			s
	$T(OS)$		108	Figure 5				10		10	80		V/µs
	$SR(+)$	4002	109	Figure 5, A_V = 1, V_{IN} = -5 V to +5 V	0.3		0.3		0.05		80		V/µs
	$SR(-)$		110	Figure 5, A_V = 1, V_{IN} = +5 V to -5 V	0.3		0.3		0.05		80		
	CS 2/		111	Figure 6	80		80		80				dB
	$BI(BB)$		112	BW = 5 kHz figure 10			15		15		15		µV rms
	$BI(PC)$		113	BW = 5 kHz figure 10			40		80		40		µV pk

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Conditions $+V_{CC}$ = +20 V dc, figure 4, unless otherwise specified	1/ -01, -02, -05 Limits		1/ -03, -08 Limits		1/ -04, -06 Limits		1/ -07 Limits		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
8 T_A = +125°C	TR(t_r)		114	T_A = +125°C, figure 5		800		800		1,000		40	ns
	TR(OS)		115	T_A = +125°C, figure 5		25		25		50		50	%
	SR(+)	4002	116	T_A = +125°C, figure 5, Av = 1, V_{IN} = -5 V to +5 V	0.3		0.3		0.05		40		V/µs
	SR(-)		117	T_A = +125°C, figure 5, Av = 1, V_{IN} = +5 V to -5 V	0.3		0.3		0.05		40		V/µs
9 T_A = -55°C	TR(t_r)		118	T_A = -55°C, figure 5		800		800		1,000		40	ns
	TR(OS)		119	T_A = -55°C, figure 5		25		25		50		50	%
	SR(+)	4002	120	T_A = -55°C, figure 5, Av = 1, V_{IN} = -5 V to +5 V	0.3		0.2		0.05		50		V/µs
	SR(-)		121	T_A = -55°C, figure 5, Av = 1, V_{IN} = +5 V to -5 V	0.3		0.2		0.05		50		V/µs
12 T_A = +25°C	t_s(+)		122	T_A = +25°C, figure 5								800	ns
	t_s(-)		123	T_A = +25°C, figure 5								800	ns
13 T_A = +125°C	t_s(+)		124	T_A = +125°C, figure 5								1,200	ns
	t_s(-)		125	T_A = +125°C, figure 5								1,200	ns
13 T_A = -55°C	t_s(+)		126	T_A = -55°C, figure 5								1,200	ns
	t_s(-)		127	T_A = -55°C, figure 5								1,200	ns

NOTES:

1/ Limits apply to both halves of dual devices (02, 05, 06, and 08) independently, and slew rate limit for device 08 is 0.4 V/µs at all temperatures (tests 109, 110, 116, 117, 120, 121).

2/ V_{CM} is achieved by algebraically subtracting the common mode voltage from each supply and algebraically adding the common mode voltage to V (i.e., for V_{CM} = -15 V, $+V_{CC}$ = +35 V, $-V_{CC}$ = -5 V, V = -15 V).

3/ V_{ID}(ADJ) is not performed on device type 02, case I only, or on device type 08 either case.

4/ Due to the significant power dissipation and associated device heating, these tests shall always be the last tests performed in any given sequence, followed by operational verification (e.g., such tests as V_{OPP}, Avs, TR, SR).

5/ Limit shown applied to single devices (-01, -03, and -04) only. The maximum quiescent I_{CC} for dual devices (-02, -05, -06, and -08) is twice that shown for single devices.

6/ Tests 29, 34, 55, and 60 which require a read and record measurement plus a calculation, may be omitted except when subgroups 2 and 3 are being accomplished for group A sampling inspection and groups C and D endpoint measurements.

7/ Applies to device types -02, -05, and -08 only.

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TABLE IV. Group C end point electrical parameters.
(T_A = +25°C, V_{CC} = ±20 V, V_{CM} = 0 V)

Table III test no.	Test	Unit	-01, -02, -08				-03, -05			
			Limit		Delta		Limit		Delta	
			Min	Max	Min	Max	Min	Max	Min	Max
3	V_{IO}	mV	-3.0	+3.0	-0.5	+0.5	-2.0	+2.0	-0.5	+0.5
11	$+I_{IB}$	nA	+1.0	+110	-12	+12	+1.0	+75	-7.5	+7.5
15	$-I_{IB}$	nA	+1.0	+110	-12	+12	+1.0	+75	-7.5	+7.5

Table III test no.	Test	Unit	-04, -06				-07			
			Limit		Delta		Limit		Delta	
			Min	Max	Min	Max	Min	Max	Min	Max
3	V_{IO}	mV	-0.5	+0.5	-0.25	+0.25	-4.0	+4.0	-1.0	+1.0
11	$+I_{IB}$	nA	-0.1	+2.0	-0.5	+0.5	+1.0	250	-25	+25
15	$-I_{IB}$	nA	-0.1	+2.0	-0.5	+0.5	+1.0	250	-25	+25

4.5 Methods of inspection. The methods of inspection shall be specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedure. When devices are measured at +25°C following application of the steady state life or burn-in condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification of the qualifying activity, if applicable.

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e. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	741A
02	747A (with common $+V_{CC}$)
03	LM101A
04	LM108A
05	LH2101A
06	LH2108A
07	LM118
08	1558

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG
Air Force - 80

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
3b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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